Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/12/2010
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	A Quantitative Test of On-Orbit Exercise Countermeasures for Bone Demineralization Using A Bedrest Analog - NSBRI		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	206-221-2845
Organization Name:	University of Washington		
PI Address 1:	Department of Orthopaedics and Sports Medicine		
PI Address 2:	School of Medicine		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	09/01/2005	**End Date:**	12/31/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2009 (from 8/31/2009) per NSBRI (1/09)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-BL00408		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bedrest is an excellent analog for the study of on-orbit musculoskeletal changes that occur during long-duration spaceflight. Our novel approach involved quantifying typical daily loads to the lower extremities in selected free-living volunteer subjects who are moderate exercisers and planning individual exercise prescriptions based on DLS theory. Subjects were randomized to control or exercise groups and confined to 6-degree head-down bedrest in the General Clinical Research Center (GCRC) at the Cleveland Clinic for 12 weeks. During this time, the exercise group underwent individualized daily exercise programs in the unique Zero Gravity Locomotion Simulator (ZLS) at CCF designed to replace their daily mechanical load stimulus experienced during free living. DXA, MRI, and QCT scans were conducted at the start and end of bedrest to assess variations in bone mineral density, changes in muscle volume, and regional changes (cortical vs. trabecular) in bone. Strength measurements were collected pre- and post-bedrest, and urinary and serum markers of bone resorption and formation were also assessed. Dietary intake sufficient to maintain constant body weight with a balanced intake of macro- and micronutrients was planned and supervised by a registered dietician, and a physician was available 24 hr/day to attend to any medical needs. A rehabilitation program was offered to subjects in the 8 weeks following bedrest. The proposed experiment should provide a categorical answer to the question of whether intermittent load replacement can adequately protect the musculoskeletal system against hypokinetic osteopenia and muscle atrophy and may open a new era in individualized exercise countermeasure planning. This study has been relocated to the NASA Flight Analogs Research Unit (FARU) at the University of Texas Medical Branch (UTMB) in Galveston, TX for completion. All data reported here were collected at the Cleveland Clinic.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained as a result of this study will provide further insight into the role of skeletal loading in bone loss during long-duration spaceflight, and knowledge gained from this research will thereby contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	The total number of subjects who have completed the study at the Cleveland Clinic is now 11 (6 controls and 5 exercisers). During the last year, the administrative and regulatory procedures needed to transfer this study to the NASA Flight Analogs Research Unit (FARU) have been completed. The study has been successfully relocated to the NASA Flight Analogs Research Unit at the University of Texas Medical Branch in Galveston, TX for completion of the remaining subjects. It is anticipated that the first subjects in the Texas arm of the study will begin bedrest in early 2010. The project team has modified existing mathematical theories on daily load stimulus (DLS) incorporating new theories on saturation and recovery of the osteogenic potential of bone with distributed mechanical loading and the effects of standing. This revised algorithm has been dubbed the enhanced DLS (EDLS). Extensive MATLAB code has been written to allow users to analyze complete sets of in-shoe force data and calculate EDLS. The EDLS is being used to formulate the individualized exercise prescriptions used by the exercising subjects. The EDLS algorithm has been documented and published in the journal "Aviation, Space, and Environmental Medicine."
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Cavanagh PR, Licata AA, Rice AJ, Gopalakrishnan R, Novotny SC, Kuklis MM, Davis BL, Genc KO, Englehaupt RK, Rizk RY, Ilaslan H. "Exercise as a countermeasure to bone demineralization during 12-weeks of bedrest." NASA Human Research Program Investigators' Workshop, League City, TX, February 2-4, 2009. Proceedings from NASA Human Research Program Investigators' Workshop, February 2009. , Feb-2009
Abstracts for Journals and Proceedings	Cavanagh PR, Rice AJ, Novotny SC, Hanson AM, Gopalakrishnan R, Kuklis MM, Davis BL, Genc KO, Englehaupt RK, Rizk RY, Ilaslan H, Licata A. "Exercise as a countermeasure to bone demineralization during 12-weeks of bedrest." 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3 - 5, 2010. Proceedings from the 2010 NASA Human Research Program Investigators' Workshop, February 2010. , Feb-2010
Abstracts for Journals and Proceedings	Genc KO, Gopalakrishnan R, Kuklis MM, Maender CC, Rice AJ, Cavanagh, PR. "Using the enhanced daily load stimulus model to quantify the mechanical load and bone mineral density changes experienced by crew members on the international space station." 56th Annual Meeting of the Orthopaedic Research Society, New Orleans, LA, March 6-9, 2010. Proceedings from the 56th Annual Meeting of the Orthopaedic Research Society, March 2010. , Mar-2010
Abstracts for Journals and Proceedings	Genc KO, Humphreys BT, Rice AJ, Englehaupt RK, Novotny SC, Gopalakrishnan R, Ilaslan H, Licata AA, Cavanagh PR. "Validating the enhanced daily load stimulus model using the bedrest analog of spaceflight." The 33rd Annual Meeting of the American Society of Biomechanics, University Park, PA, August 26-29, 2009. Proceedings from the 33rd Annual Meeting of the American Society of Biomechanics, August 2009. , Aug-2009
Articles in Peer-reviewed Journals	Genc KO, Humphreys BT, Cavanagh PR. "Enhanced daily load stimulus to bone in spaceflight and on earth." Aviat Space Environ Med. 2009 Nov;80(11):919-26. PubMed PMID: 19911514 , Nov-2009
Articles in Peer-reviewed Journals	Cavanagh PR, Rice AJ, Novotny SC, Genc KO, Englehaupt RK, Owings TM, Comstock B, Cardoso T, Ilaslan H, Smith SM, Licata AA. "Replacement of daily load attenuates but does not prevent changes to the musculoskeletal system during bed rest." Bone Reports. 2016 Dec;5:299-307. http://dx.doi.org/10.1016/j.bonr.2016.10.001 , Dec-2016